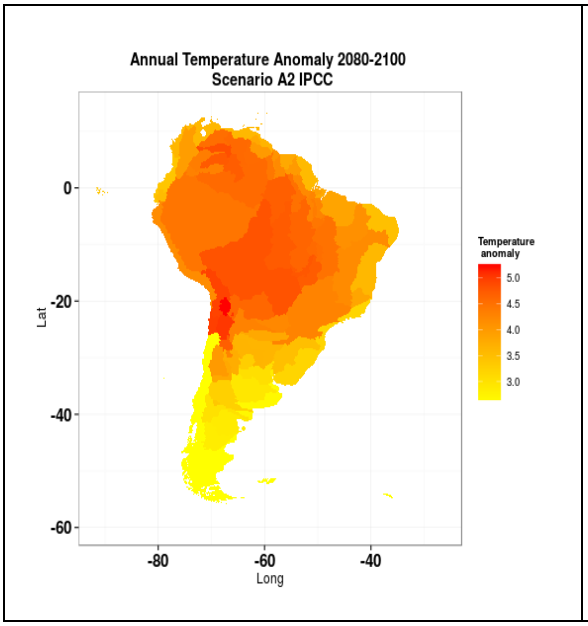
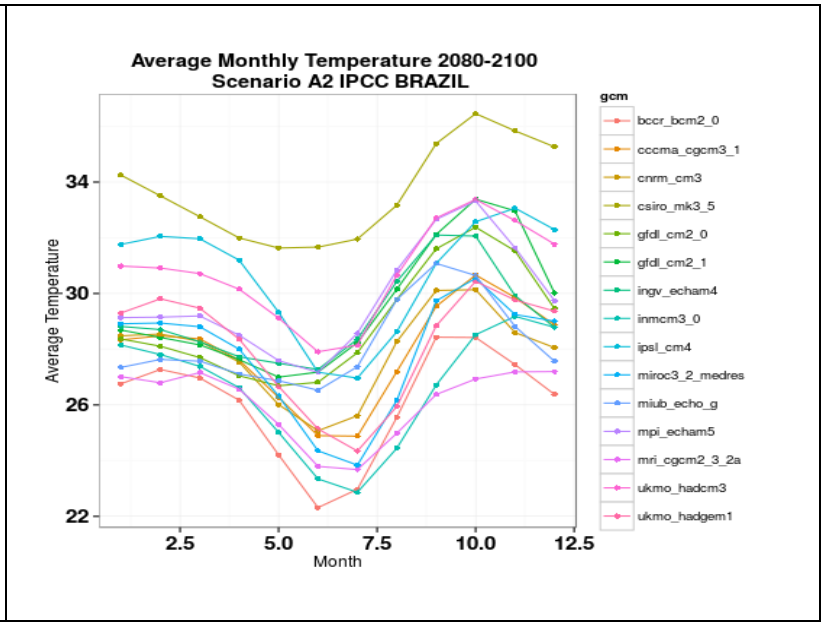
ANALYZING IPCC GLOBAL CLIMATE MODELS WITH rWBclimate IN SOUTHAMERICA

S. Ibarra (1), A. Rehbein (1)
(1) Department of Atmospheric Sciences, Institute of Astronomy, Geophysics and Atmospheric Sciences, University of São Paulo, Brazil
Presenting author email: sibarra@iag.usp.br

The new rWBclimate R Package allow make analyzes of IPCC global climate models. Data of the IPCC Fourth Assessment Report: Climate Change 2007 (AR4) [1] were analyzed for Southamerica. The scenario A2, which follows an increase in population and economic growth was evaluated using the percentile 50 of the Global Climate Models [2] of annual temperature anomaly for 2080 and 2100. The analyses was made using the software R [3], and the packages rWBclimate [4] and ggplot2 [5]. The results shows temperature anomalies between 3 ºC and 5 ºC with the highest value in north zone of Chile, close to Atacama dessert. All the zone near equatorial line shows an increase of around 4 ºC. The amazonian regions shows higher temperature anomaly increase. When analyzing the average monthly temperature for Brazil for the same period of years, the minimum temperature is 22.31391 ºC with the bccr_bcm2_0 [6] model, and the maximum is 36.45606 ºC with the csiro_mk3_5 model [7]. The rWBclimate package allows reproducibility and fast analysis.





1. Pachauri, R. K., & Reisinger, A. (2007). IPCC fourth assessment report. IPCC, Geneva.
2. http://www.ipcc-data.org/sim/gcm_monthly/SRES_AR4/index.html
3. R Core Team (2014). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL http://www.R-project.org/.
4. Edmund Hart (). rWBclimate: A package for accessing World Bank climate data. R package version 0.1.3 http://github.com/ropensci/rWBclimate
5. H. Wickham. ggplot2: elegant graphics for data analysis. Springer New York, 2009.
6. Furevik, Tore, et al. "Description and evaluation of the Bergen climate model: ARPEGE coupled with MICOM." Climate Dynamics 21.1 (2003): 27-51.
7. The CSIRO Mk3 climate system model. Aspendale, Victoria, Australia: CSIRO Atmospheric Research, 2002.